1 **(a)** Given that a is an integer, show that the system of equations

$$\begin{aligned} ax + 3y + z &= 14, \\ 2x + y + 3z &= 0, \\ -x + 2y - 5z &= 17, \end{aligned}$$

has a unique solution and interpret this situation geometrically. [4]

(b) Find the value of a for which $x = 1,\ y = 4,\ z = -2$ is the solution to the system of equations in part **(a)**. [1]

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